

$$ds^2 = -dt^2 + a^2(t) \left(\frac{dr^2}{1-Kr^2} + r^2 d\Omega^2 \right)$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G_N T_{\mu\nu}$$

$$T_{\mu\nu} = \begin{pmatrix} \rho & & \\ & p & \\ & & p \end{pmatrix}$$

$\Rightarrow$ why is the energy momentum a (!) tensor?

$$\rho = \rho_r + \rho_m + \rho_\Lambda$$

$$\begin{matrix} \Downarrow & \Downarrow & \Downarrow \\ p = \frac{1}{3}\rho & p = 0 & p = -\rho \end{matrix}$$

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi}{3} G_N \rho - \frac{K}{a^2} \quad (1)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi}{3} G_N (\rho + 3p) \quad (2)$$

$$F = \frac{GMm}{r^2}$$

$$\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = \frac{\text{G} \cdot \text{kg}^2}{\text{m}^2}$$

Setting $K=0$

$$G = \frac{\text{m}^4}{\text{s}^2 \text{kg}} \quad \text{FI} \Rightarrow$$

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi}{3} G_N \rho \Rightarrow \left| \rho_{\text{crit}} = \frac{H_0^2}{\frac{8\pi}{3} G_N} \right|$$

$$\Omega_{i0} = \frac{\rho_{i0}}{\rho_{\text{crit}}}$$

$$0 = -dt^2 + a^2(t) \frac{dr^2}{1 - Kr^2}$$

$$\frac{dt}{a(t)} = \frac{dr}{\sqrt{1 - Kr^2}}$$

$$\int_{t_{\text{emission}}}^{t_{\text{observed}}} \frac{dt}{a(t)} = \int_{r_e}^{r_o} \frac{dr}{\sqrt{1 - Kr^2}}$$

$$|n|a(t)| =$$

$$\int_{t_e}^{t_o + \delta t_o} \frac{dt}{a(t)} \approx \int_{t_e}^{t_o} \frac{dt}{a(t)} + \left[\frac{\delta t_o}{a(t_o)} - \frac{\delta t_e}{a(t_e)} \right]$$

Why can you do this? It's allowed?

2 beams of light shot at very close time intervals.

Hence:

$$\frac{\delta t_o}{a(t_o)} = \frac{\delta t_e}{a(t_e)}$$

$$\Rightarrow \frac{\delta t_o}{\delta t_e} = \frac{a(t_o)}{a(t_e)} = \frac{v_e}{v_o} = \frac{\lambda_o}{\lambda_e}$$

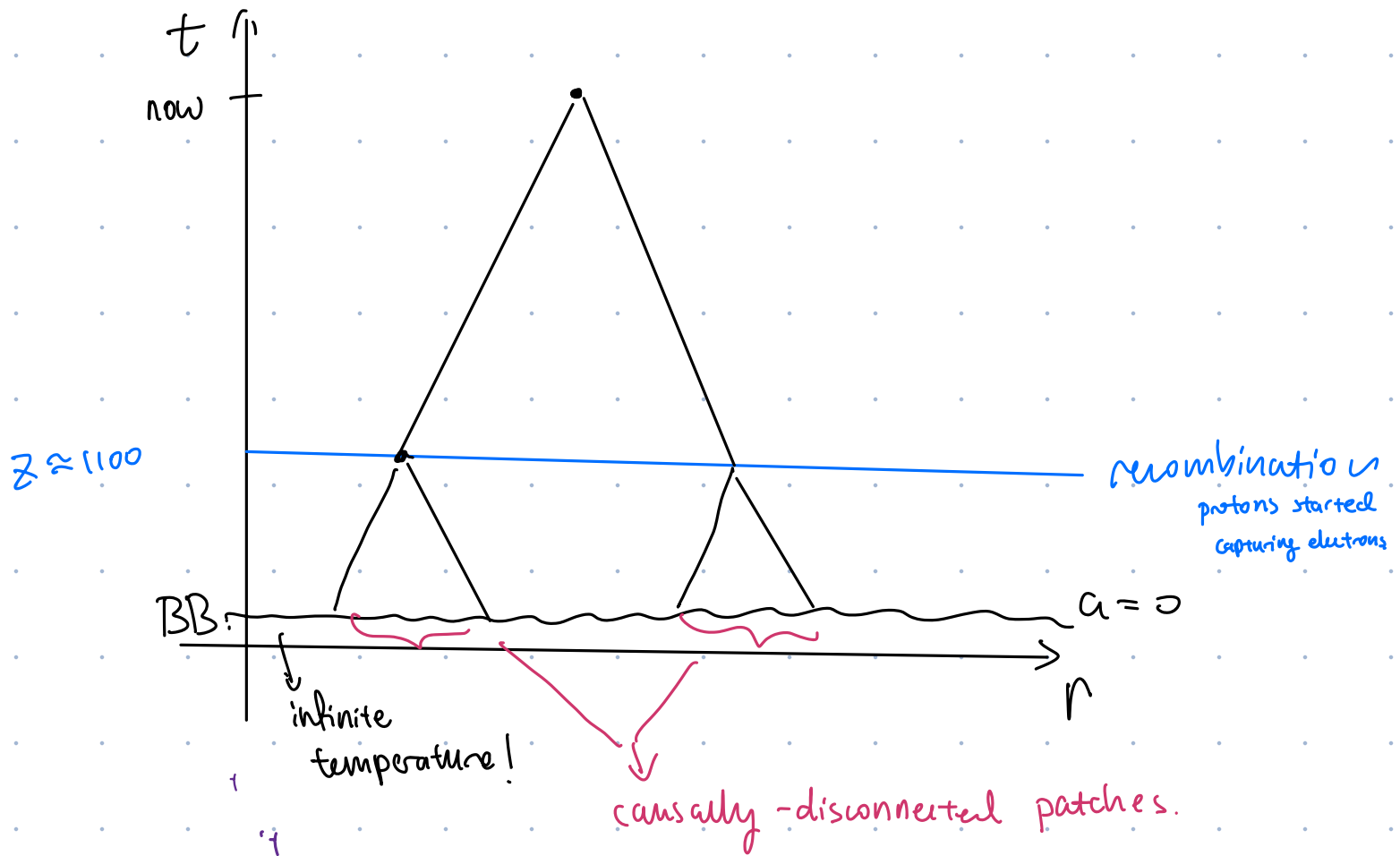
how red-shifted the light is when they observe it.

$$z \equiv \frac{\lambda_{\text{obs}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} - 1$$

$$= \frac{a(t_o)}{a(t_e)} - 1$$

$$= \frac{1}{a(t_e)} - 1$$

$$a(t_o) = \frac{1}{z+1}$$



How big
were the patches
for $a=10^{-32}$?

why do causally-disconnected patches
have the same temperature?

→ The Horizon Problem.

$$0 = -dt^2 + a^2(t)dr^2 \quad K=0.$$

$$\frac{dt}{dr} = a \sim t^q \rightarrow q = \frac{2}{3} \text{ radiation}$$

$\frac{1}{2}$ matter.

$$ds^2 = -dt^2 + a(t) \left(\frac{dr^2}{1-Kr^2} + r^2 d\Omega^2 \right)$$

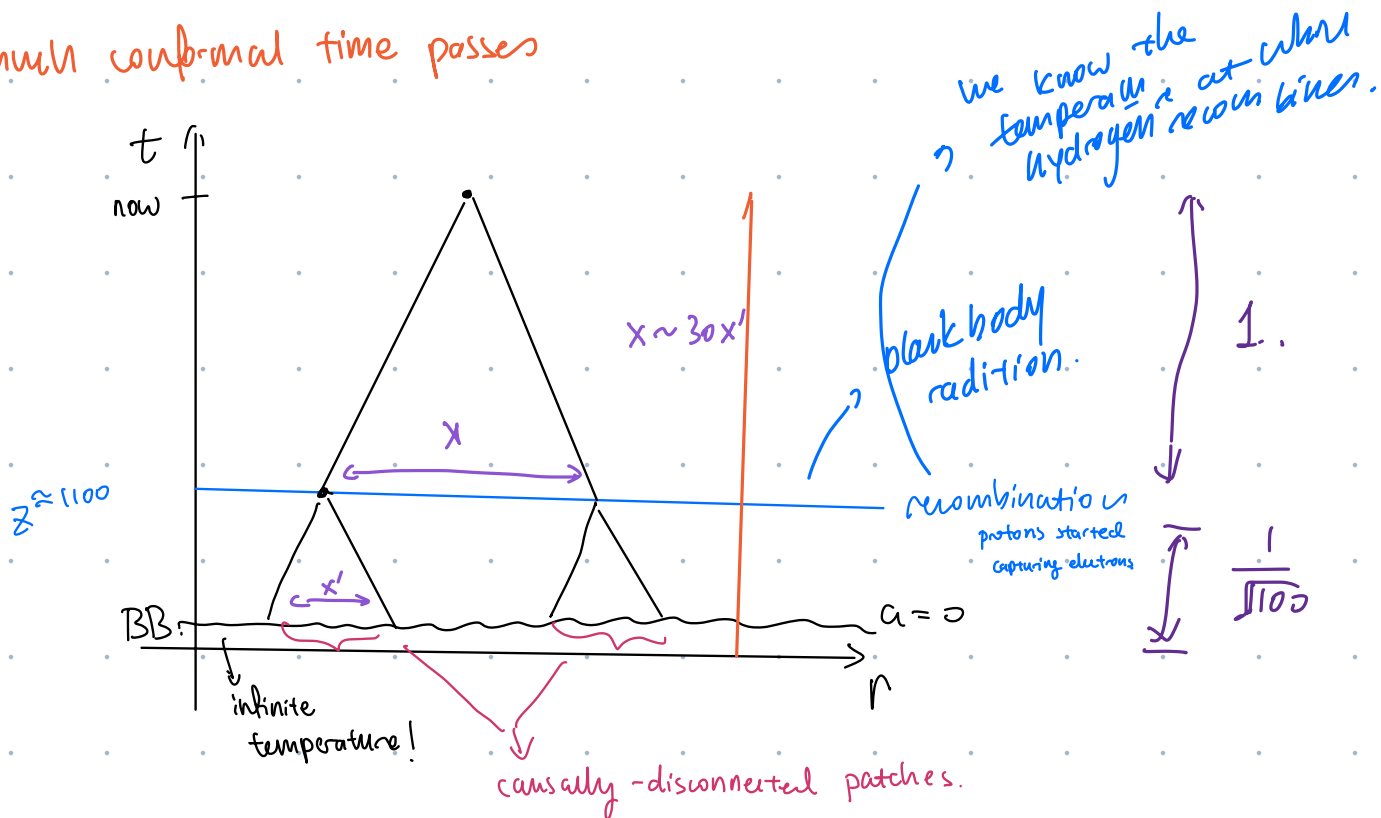
$$= a^2(t) \left[-\frac{dt^2}{a^2(t)} + \frac{dr^2}{1-Kr^2} + r^2 d\Omega^2 \right]$$

← conformal.

$$dt^c = \frac{dt}{a(t)}$$

$$ds^2 = a^2(t) \left[-(dt^c)^2 + \frac{dr^2}{1-Kr^2} + r^2 d\Omega^2 \right]$$

How much conformal time passes

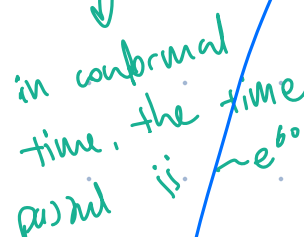


$$t^c = \int \frac{dt}{a(t)} = \int \frac{dt}{t^{2/3}} = \int t^{-2/3} dt$$

↑ matter

$$= 3 t^{1/3} \Big|_{t_1}^{t_2} \approx \sqrt{a}$$

$a \propto \left(\frac{t}{t_{\text{now}}} \right)^{2/3}$

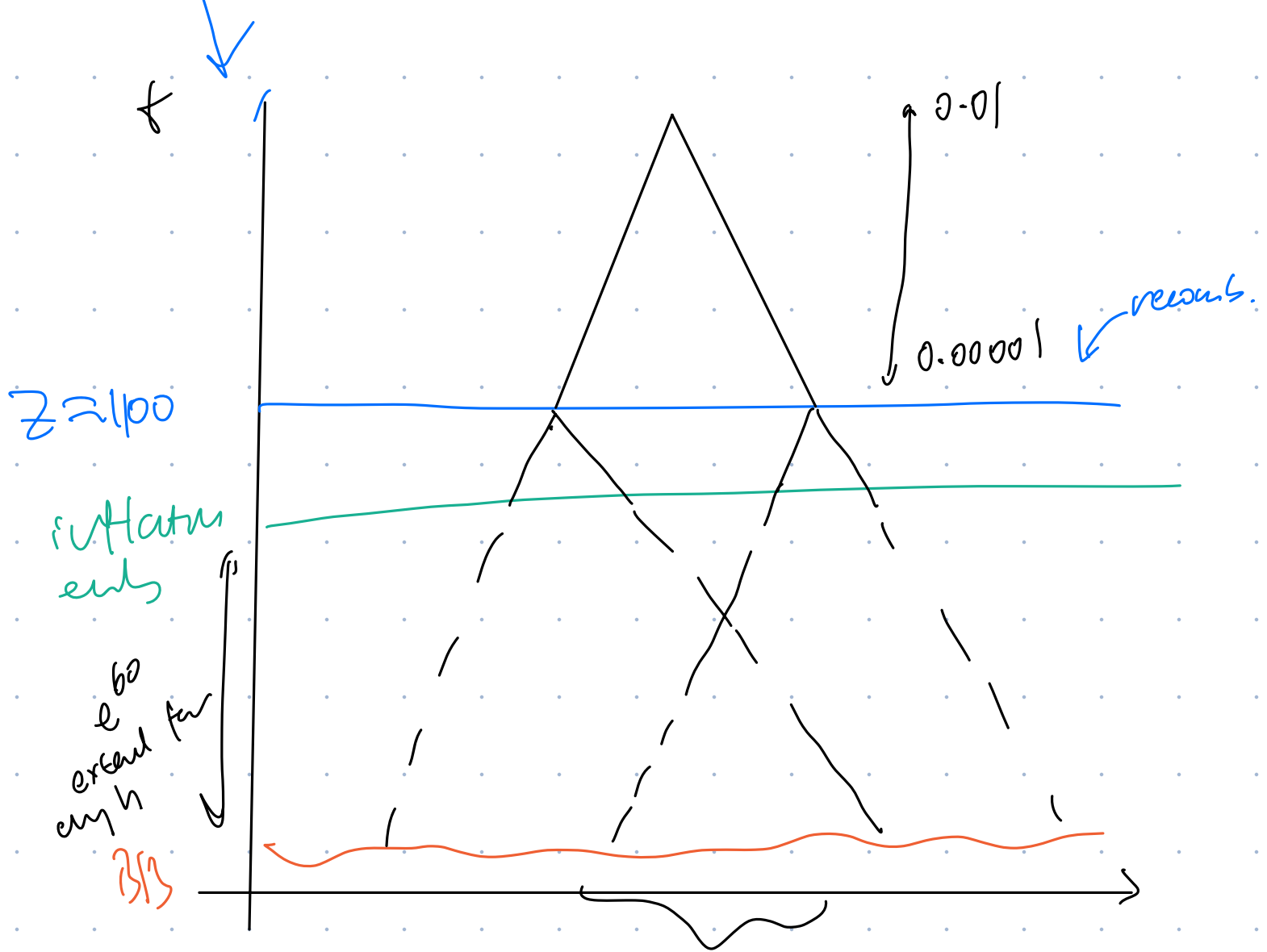


$$\frac{dC}{dt} = \frac{dI}{A e^{\lambda t}}$$

$$t_{\text{End}}^{\text{C}} - t_{\text{reg}}^{\text{C}} = -\frac{1}{A\lambda} (e^{-\lambda t_E} - e^{-\lambda t_B})$$

$$= \frac{e^{-\lambda t_E}}{A\lambda} \left(e^{-\lambda(t_B - t_E)} - 1 \right)$$

$$= \frac{1}{A \lambda e^{\lambda E}} [e^{-\lambda(t_B - t_E)} - 1] \geq e^{-\lambda t_0}$$



you are looking at the same patch of spacetime.

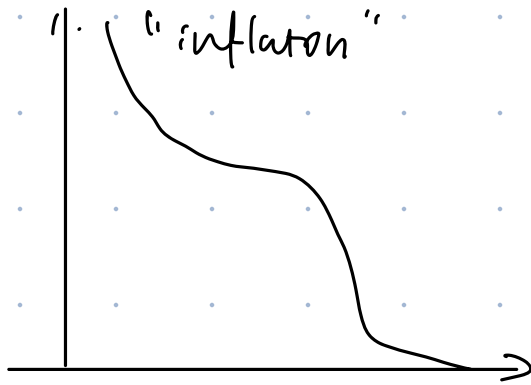
The flatness

Certain theories that produce magnetic monopoles.

We don't know how this happens!

↓
how does the inflation stop?

We don't have a convincing
physical picture of
the Cosmic
Inflation Theory.



$$w = -1 \quad , \quad P = -\rho$$

↓ ↖ what → this?

Nikhil:

w changes with
time.

↓
but experiments are
incomparable.

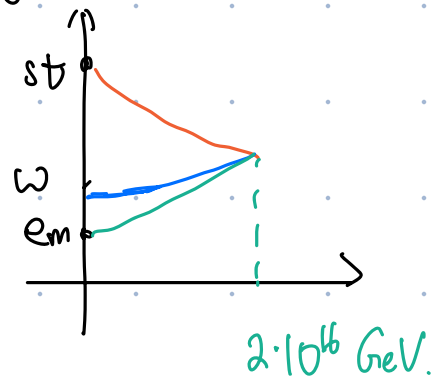
$$\Omega_m \approx 0.3 \quad \sim 0.05 \rightarrow \text{baryonic}$$

↑
dark matter.

Monopoles are predicted by any Grand Unified Theories.

↳ Predictions:
Protons decay

Coupling constants.



No.
Solid experimental
evidence?

why aren't electric ~~mon~~ monopoles diluted?